

Diego Alejandro Salazar

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Professional Summary

Gameplay Programmer specialized in **Unity (C#)** and **Unreal Engine 5 (C++)** with a focus on modular gameplay systems, **Software Architecture**, and performance optimization. Experienced in **Agile** methodologies, AI integration (Whisper), Niagara VFX, and 2D Vector Asset Design. Committed to writing clean, scalable code for immersive player experiences.

Technical Skills & Languages

Languages: C#, C++, C, TypeScript, SQL, Matlab

Software: Git, Visual Studio, Adobe Illustrator

Core Skills: Gameplay Systems, Niagara VFX, Audio DSP

Languages: Italian (Native), English (B2), Spanish (Conv.)

Engines: Unity, Unreal Engine 5, JUCE

Design: OOP, Design Patterns, State Machines

Art/UI: 2D Vector Art, Prototyping, Pinterest

Education

BSc in Information Engineering for Video Games and Virtual Reality

2022–2025

Polytechnic University of Marche (UNIVPM)

- **Core Subjects:** Advanced Game Programming, Data Structures & Algorithms, UX Design for Games, Circuits & Sound Design, Digital Signal Processing, Database Management.

High School Diploma | IIS A. Volta Pescara

2017–2022

Project Thesis and Apprenticeship

Unilingo: AI-Driven Language Learning Game

Unity, C#, Whisper, TypeScript

Lead Programmer and Data Collection Lead

- Developed a privacy-first Japanese learning game using **Local AI models** for speech recognition and image processing.
- Engineered a data pipeline using a **TypeScript script** to parse JSON gameplay logs into analytical datasets.

Selected Game Development Projects

3D Platformer Prototype (Team Manager & Core Programmer)

Unreal Engine 5, C++

Project Management & Systems Architecture

- Programmed all **core gameplay systems** and custom **Niagara VFX** (excluding animations/UI design).
- Managed project lifecycle using **Agile/Scrum** principles and performed code reviews.

Educational Game (Programmer & Lead Artist)

Unity, C#, Adobe Illustrator

Gameplay Design and Asset Creation

- Designed and created all **2D vector assets** using **Adobe Illustrator**, following moodboards to ensure a cohesive visual style.
- Implemented gameplay systems and UI flow for sensitive educational content in collaboration with pedagogical advisors.

Current Project: PS1-style Action RPG

Unity, C#, Lead Programmer

Systems Programming & Optimization

- Implementing Action Combat, Inventory, Dialogue and Quest systems with **Additive Scene Streaming** and **IK Animations**.

Text-Based RPG Adventure (Team)

C#, Systems Design

- Architected a branching narrative system, turn-based **combat system**, and persistent save/load logic.

Audio Plugin: DSP Distortion, Delay & Reverb

JUCE, C++, Audio DSP

- Designed an interactive UI and connected parameters via `AudioProcessorValueTreeState`.

First-Person Horror/Puzzle & Game Jam Projects

Unity, C#

- Optimized performance through **Object Pooling** and scene loading; delivered 3D prototypes in a few time.

NAO Challenge (Team)

Robotics / C++

- Scripted interactive behaviors and state-based dialogue for robotic hardware.